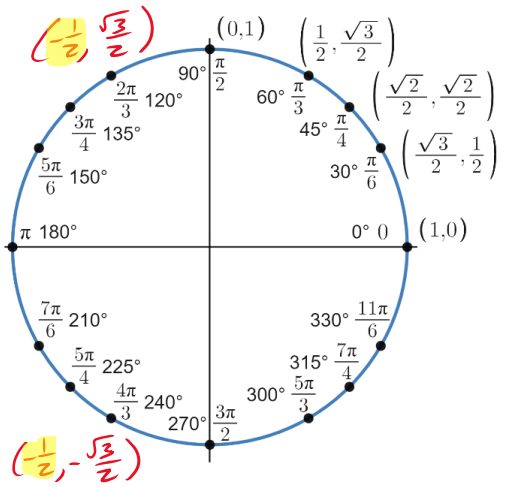
Solving Trigonometric Equations

Solving Trigonometric Equations by Graphing

Perhaps the best way to introduce solutions to trigonometric equations is to view them from a graphing perspective. Both the unit circle (circular graph) and the rectangular graphs are helpful when solving these equations.

EXAMPLE 1

Use the provided graphs to help solve each equati



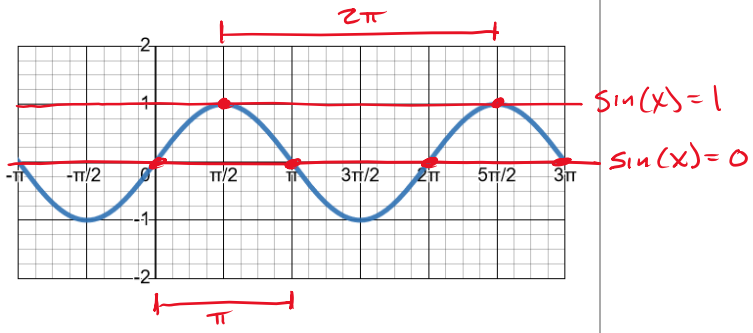
(a) $\sin(x) = 1$

$X = \frac{\pi}{2} + 2\pi k$, k any integer

(b) $\sin(x) = 0$

$X = 0 + \pi k$

$X = \pi k$, k any integer



(c) $2\cos(x) + 1 = 0$

$\cos(X) = -\frac{1}{2}$

$X = \frac{2\pi}{3} + 2\pi k$, k any integer

$X = \frac{4\pi}{3} + 2\pi k$, k any integer

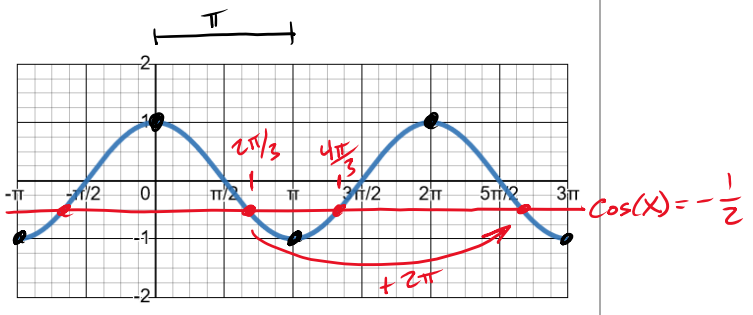
(d) $\cos^2(x) = 1$

$\cos(X) = \pm\sqrt{1}$

$\cos(X) = \pm 1$

$X = 0 + \pi k$

$X = \pi k$, k any integer

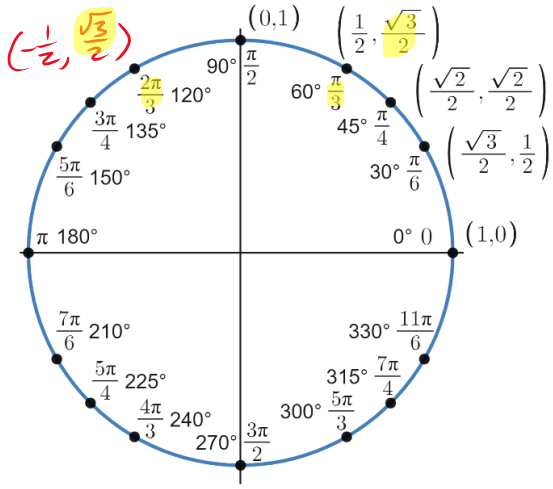


Linear Methods

We can apply the familiar techniques for solving linear equations when solving trigonometric equations. The goal is to isolate the trigonometric function using algebraic operations such as adding, subtracting, multiplying, and dividing on both sides.

EXAMPLE 2

Solve each equation. Give solutions on the interval $[0,2\pi)$ and again on the interval $(-\infty,\infty)$.



(a) $3\sqrt{3}\csc(x) = 6$

$\csc(x) = \frac{6}{3\sqrt{3}} = \frac{2}{\sqrt{3}}$

$\sin(x) = \frac{\sqrt{3}}{2}$, $x = \{\frac{\pi}{3}, \frac{2\pi}{3}\}$

$x = \frac{\pi}{3} + 2\pi k$

$x = \frac{2\pi}{3} + 2\pi k$

(b) $3\sqrt{3}\csc(2x) = 6$

$\sin(2x) = \frac{\sqrt{3}}{2}$

$\sin(\theta) = \frac{\sqrt{3}}{2}$, $\theta = \frac{\pi}{3} + 2\pi k$ $\quad$ $\theta = \frac{2\pi}{3} + 2\pi k$

$2x = \frac{\pi}{3} + 2\pi k$ $\quad$ $2x = \frac{2\pi}{3} + 2\pi k$

$x = \frac{\pi}{6} + \pi k$ $\quad$ $x = \frac{\pi}{3} + \pi k$

$x = \{\frac{\pi}{6}, \frac{7\pi}{6}, \frac{\pi}{3}, \frac{4\pi}{3}\}$

(c) $2\tan\left(\frac{x}{2}\right) - 1 = \tan\left(\frac{x}{2}\right)$

$\tan(\frac{x}{2}) = 1$

$\tan(\theta) = 1$, $\theta = \frac{\pi}{4} + \pi k$

$\frac{x}{2} = \frac{\pi}{4} + \pi k$

$x = \frac{\pi}{2} + 2\pi k$

$x = \frac{\pi}{2}$

Factoring Methods

There are times when basic arithmetic (addition, subtraction, multiplication, and division) is not sufficient to isolate the trigonometric function. In these cases, factoring methods and the zero-product property may be helpful. Recall the zero-product property states if $a \cdot b = 0$, then either $a = 0$ or $b = 0$ (or both).

EXAMPLE 3

Use factoring and the zero-product property to solve each trigonometric equation. Give solutions on the interval $[0°, 360°)$ and again on $(-\infty, \infty)$

(a) $\sin^2(x) = 3\sin(x)$

$\sin^2(x) - 3\sin(x) = 0$

$\sin(x)(\sin(x) - 3) = 0$

$\sin(x) = 0$

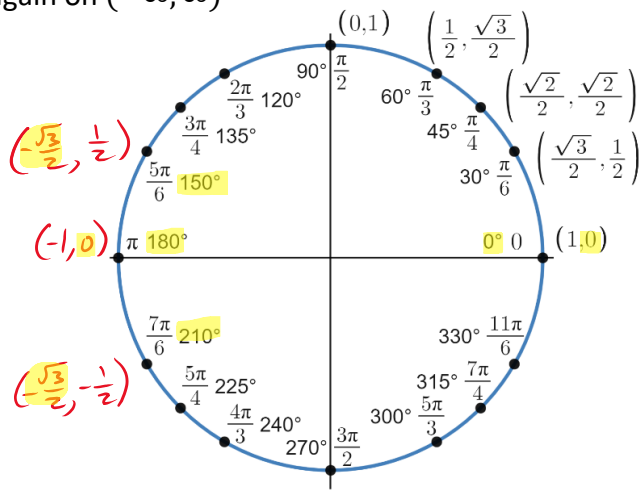
$x = 0° + 180°k$, k any integer

$x = \{0°, 180°\}$

$\sin(x) - 3 = 0$

$\sin(x) = 3$

no soln



(b) $\sin(x)\sec(x) = -\frac{2}{\sqrt{3}}\sin(x)$

$\sin(x)\sec(x) + \frac{2}{\sqrt{3}}\sin(x) = 0$

$\sin(x)\left(\sec(x) + \frac{2}{\sqrt{3}}\right) = 0$

$\sin(x) = 0$

$x = 0° + 180°k$, k any integer

$\sec(x) + \frac{2}{\sqrt{3}} = 0$

$\sec(x) = -\frac{2}{\sqrt{3}}$

$\cos(x) = -\frac{\sqrt{3}}{2}$

$x = 150° + 360°k$, k any integer

$x = 210° + 360°k$, k any integer

$x = \{0°, 180°, 150°, 210°\}$

EXAMPLE 4

Use factoring and the zero-product property to solve each trigonometric equation. Give solutions on the interval $[0,2\pi)$.

$2\cos^2(x) - 3\cos(x) + 1 = 0$

let $u = \cos(x)$

$2u^2 - 3u + 1 = 0$

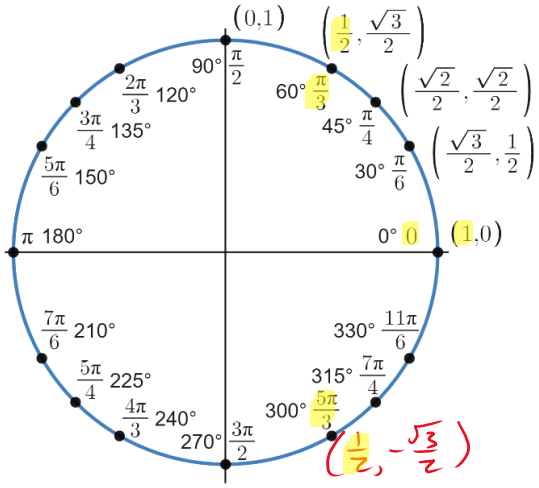
$(2u-1)(u-1) = 0$

$u = \frac{1}{2}$ $\qquad$ $u = 1$

$\cos(x) = \frac{1}{2}$ $\qquad$ $\cos(x) = 1$

$x = \frac{\pi}{3}, x = \frac{5\pi}{3}$ $\qquad$ $x = 0$

$x = \left\{\frac{\pi}{3}, \frac{5\pi}{3}, 0\right\}$



Applying Identities

When dealing with equations that involve more than one trigonometric function, or multiple angles, it may be helpful to apply an identity and rewrite the equation in terms of a single trigonometric function or a single angle.

EXAMPLE 5

Find a useful identity and solve the equation. Give solutions on the interval $(-\infty, \infty)$.

$3\sin(2x) = -2\sin(x)\cos(x)$, multiple angles!

$3\sin(2x) = -\sin(2x)$, one angle

$4\sin(2x) = 0$

$\sin(2x) = 0$

$2x = 0 + \pi k$

$x = \frac{0}{2} + \frac{\pi k}{2}$

$x = \frac{\pi k}{2}$, k any integer

EXAMPLE 6

Find a useful identity and solve the equation. Give solutions on the interval $(-\infty, \infty)$.

$-2\sin^2(x) - 3\cos(x) + 3 = 0$, mult. trig fcts

$-2(1 - \cos^2(x)) - 3\cos(x) + 3 = 0$, one trig fct

$2\cos^2(x) - 3\cos(x) + 1 = 0$

$x = \{0, \frac{\pi}{3}, \frac{5\pi}{3}\}$ (from ex 4)

$x = 0 + 2\pi k$

$x = \frac{\pi}{3} + 2\pi k$

$x = \frac{5\pi}{3} + 2\pi k$

Calculator Approximations

We will not always work exclusively with values found along the unit circle. In applications, calculator approximations are frequently required to obtain results. More specifically, we often need the calculator to approximate inverse trigonometric function values. Two identities are particularly helpful in these cases.

IMPORTANT INFO: ***Useful Identities for Solving Trigonometric Equations***	
For trigonometric equations requiring calculator approximations, the inverse functions are often needed. Due to the restrictions placed on inverse trigonometric functions, it may be convenient to apply the following identifies to ensure that all solutions are found.	
Even Identity for Cosine	$\cos(-\theta) = \cos(\theta)$
Supplementary Angle Identity for Sine *(a result of the difference identity for sine)*	$\sin(\theta) = \sin(\pi - \theta)$, θ in radians $\sin(\theta) = \sin(180° - \theta)$, θ in degrees

EXAMPLE 7

Solve each equation and give solutions on the interval $[0,2\pi)$. → $[0, 6.28)$

(a) $2\sec(\theta) = 3$

$\sec(\theta) = \frac{3}{2}$

$\cos(\theta) = \frac{2}{3}$

$\theta = \cos^{-1}\left(\frac{2}{3}\right) \approx 0.841$

AND $\theta \approx -0.841 \rightarrow \theta = -\cos^{-1}\left(\frac{2}{3}\right) + 2\pi \approx 5.442$

↑ does not satisfy $[0, 6.28)$

(b) $5\sin(x) + 4 = -2\sin(x)$

$7\sin(x) = -4$

$\sin(x) = -\frac{4}{7}$

$x = \sin^{-1}\left(-\frac{4}{7}\right) \approx -0.608 \rightarrow x = \sin^{-1}\left(-\frac{4}{7}\right) + 2\pi \approx 5.675$

↑ does not satisfy $[0, 6.28)$

AND $x = \pi - \sin^{-1}\left(-\frac{4}{7}\right) \approx 3.750$